

**3rd Semester cse Branch**

# ***DATA STRUCTURE AND ALGORITHMS***

**VVI SUBJECTIVE**





# LAKSHYA BATCH *FOR CSE*



Start Date:

18<sup>th</sup> May 2026

4<sup>TH</sup> SEMESTER



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## B.TECH 3<sup>RD</sup> SEM OSA

- ✓ Q.1 (a) Define Asymptotic Notations ( $O$ ,  $\Omega$ ,  $\Theta$ ). Explain the Time-Space Trade-off with a suitable example. Prove that  $3n + 2 = O(n)$
- ✓ (b) Compare Linear Data Structures and Non-Linear Data Structures. Discuss the elementary operations performed on Data structures.

$$7 \times 2 = 14$$

Qn.2 (a) Convert the following Infix expression to Postfix using a stack:  $A + (B * C - (D / E ^ F) * G) * H$ . Show the status of the stack at each step.

$7 \times 1 = 7$

✓ (b) Explain the concept of a Circular Queue. Why is it preferred over a Simple Queue?  
Write the algorithm for insertion in a Circular Queue.

✓ Q.3 (a) Write a complete C/C++ algorithm to insert a new node at a specific position in a Singly Linked List.

✓ (b) Compare Singly, Doubly, and Circular Linked Lists. Discuss the advantages of using a Header Node in a Linked List.



## B.TECH 3<sup>RD</sup> SEM DSA

✓ Q.4 (a) Sort the following array using Quick Sort: [45, 12, 89, 7, 34, 66, 10, 55]. Show all partitions and the pivot selection process.

✓ (b) Define Hashing. Explain the Collision Resolution techniques: Linear Probing and Chaining with examples.

✓ (c) Write the Merge Sort algorithm. Discuss its time complexity in Best, Average, and Worst cases.





## B.TECH 3<sup>RD</sup> SEM OSA

- ✓ Q.5 (a) Construct a Binary Search Tree (BST) for the following elements: 50, 30, 20, 40, 70, 60, 80. Perform Inorder, Preorder, and Postorder traversals.
- ✓ (b) What is an AVL Tree? Explain LL and RR rotations with a diagram.

- ✓ Q.6 (a) Differentiate between B-Trees and B+ Trees. In what scenarios is a B+ Tree preferred over a B-Tree? ✓
- ✓ (b) Explain the concept of a Threaded Binary Tree. How does it help in reducing memory overhead and improving traversal? ✓



## B.TECH 3<sup>RD</sup> SEM OSA

✓ Q.7 (a) Represent the following graph using an Adjacency Matrix and Adjacency List:  
Vertices {A, B, C, D, E}, Edges {(A, B), (B, C), (C, D), (D, E), (E, A), (A, C)}. (7)

✓ (b) Write the algorithm for Breadth First Search (BFS). Explain its application in finding the shortest path in an unweighted graph. ✓

## B.TECH 3<sup>RD</sup> SEM DSA

✓ Q.8 (a) Compare Merge Sort and Heap Sort in terms of time complexity and auxiliary space.

✓ (b) Explain Depth First Search (DFS) using a stack. Traverse a sample graph to demonstrate the process



Q.9 (a) Write an algorithm to implement a Stack using a Linked List. Why is Linked List implementation preferred over Array implementation?

✓ (b) Convert the following Infix expression to Postfix using Stack:

$$X = (A + B) * (C/D) - E$$

- ✓ Q.10 (a) Write a function/algorithm to insert a node at a specific position in a Singly Linked List.
- ✓ (b) Explain the difference between Singly, Doubly, and Circular Linked Lists with their node structures.



## B.TECH 3<sup>RD</sup> SEM DSA

✓ Q.11 Construct an AVL Tree by inserting the following elements one by one: 50, 40, 35, 30, 20, 25, 28. Show the rotations performed at each step.

✓ (b) Define B-Tree. List its properties and explain why it is widely used in Database systems.

✓ Q.12 (a) For a given Binary Tree, the In-order is: DBEAF C and Pre-order is: ABDECF .  
Construct the Binary Tree and find its Post-order traversal.

✓ (b) What is a Binary Search Tree (BST)? Write an algorithm to delete a node having two children from a BST.



✓ Q.13 Write short notes on any two of the following:

- ✓ (i) Priority Queue and its applications.
- ✓ (ii) Collision resolution techniques in Hashing.
- ✓ (iii) Heap Sort and its complexity.
- ✓ (iv) Threaded Binary Tree.

THANK YOU